

1. Administrative & Study Identification

**Full Study Title:** A Real-World Study of  $\beta$ -Thalassemia Major Treatment With Luspatercept in Taiwan  
**Official Title:** Taiwan  $\beta$ -Thalassemia Major Real-World Study for Luspatercept **Short Title/Acronym:** CA056-1132 **Protocol Number:** CA056-1132  
**NCT Number:** NCT06596642  
**Posting ID:** 734905597602233248 **Sponsor / Company Name:** Bristol Myers Squibb  
**Trial Status:** ACTIVE\_NOT\_RECRUITING (Active but not currently recruiting) **Investigational Product:** Luspatercept (Trade Name: Reblozyl, ATC Code: B03XA06) **Phase of Development:** Not Applicable  
**Disease Areas:**  $\beta$ -Thalassemia Major **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

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2. Study Synopsis & Rationale

**2.1 Study Objective Primary Objective:** To assess the efficacy and safety of luspatercept treatment for  $\beta$ -thalassemia major in a real-world clinical setting in Taiwan. **Secondary Objectives:** To evaluate the impact of luspatercept treatment on the quality of life (QoL) of patients with transfusion-dependent  $\beta$ -thalassemia major.

**2.2 Background & Rationale** To generate real-world evidence (RWE) on the effectiveness and safety of luspatercept among Taiwanese patients with  $\beta$ -thalassemia major. While clinical trials have demonstrated the efficacy of luspatercept in reducing red blood cell (RBC) transfusion burden, this observational study is conducted to understand its long-term outcomes, tolerability, and impact on patient quality of life within routine clinical practice.

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3. Study Design & Methodology

**3.1 Design Framework Study Type:** Observational (Real-World Study)  
**Control Type:** Single-arm (Cohort) **Blinding:** Open-label  
**Randomization:** N/A (Non-probability sample)

**3.2 Planned Enrollment & Duration Target** **\$\$\$:** 61 evaluable patients **Number of Sites:** Multiple sites across Taiwan (e.g., Kaohsiung Medical University Chung-Ho Memorial Hospital, National Cheng Kung University Hospital, MacKay Memorial Hospital, National Taiwan University Hospital, Chang Gung Memorial

**4. Subject Selection (Eligibility Criteria)**

**4.1 Inclusion Criteria** 1. Participants  $\geq 18$  years of age. 2. Participants with the diagnosis of transfusion-dependent  $\beta$ -thalassemia major who are eligible to the treatment of luspatercept. 3. Before luspatercept treatment, participant's transfusion burden  $\geq 24$  Red Blood Cell units in 24 weeks, without  $\geq 35$ -day transfusion free.

**4.2 Exclusion Criteria** 1. Hypersensitivity to the active substance or to any of the excipients. 2. Pregnancy. 3. Participants requiring treatment to control the growth of extramedullary hematopoiesis (EMH) masses.

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**5. Intervention & Treatment Plan**

**5.1 Investigational Regimen Dosage/Frequency:** Luspatercept administered according to the local approved label and routine clinical practice guidelines for  $\beta$ -thalassemia major. (EMA Product Information: [https://www.ema.europa.eu/en/documents/product-information/reblozyl-epar-product-information\\_en.pdf](https://www.ema.europa.eu/en/documents/product-information/reblozyl-epar-product-information_en.pdf))  
**Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** Continuous real-world treatment through the observation period (up to the study end date in October 2026).

**5.2 Concomitant & Prohibited Therapy Permitted:** Standard of care therapies for  $\beta$ -thalassemia major, including iron chelation therapy and routine RBC transfusions as clinically indicated.  
**Prohibited:** Concurrent therapies contraindicated in the local luspatercept prescribing information (e.g., targeted treatments for expanding EMH masses).

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**6. Study Timeline & Visit Schedule**

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|-------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------|---------------|
| Visit ID                                                                                                                                  | Window (Days) | Key Activities/Procedures     | :---          |
| :---                                                                                                                                      | :---          | <b>Screening / Enrollment</b> | Pre-Treatment |
| Informed Consent, Medical History, Transfusion burden assessment (verifying $\geq 24$ RBC units in the past 24 weeks)     <b>Baseline</b> |               |                               |               |
| Day 0   Initiation of luspatercept treatment (or index date), Baseline QoL assessment     <b>Interim / Follow-up</b>   Routine Care       |               |                               |               |
| Intervals   Real-world safety assessments, ongoing RBC transfusion tracking, hemoglobin monitoring     <b>End of Study</b>                |               |                               |               |
| Oct 2026   Final clinical outcome abstraction, overall safety evaluation, end-of-study QoL metrics                                        |               |                               |               |

*(Note: Exact visit schedules follow routine clinical practice rather than strict protocol-driven intervals)*

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoints** 1. Number of Red Blood Cell (RBC) transfusion units received by participants 2. Participant hemoglobin (Hb) level results 3. Participant serum ferritin results

*Measurement Method: Abstraction of clinical records, assessing transfusion frequency/volume and standard post-marketing pharmacovigilance reports.*

**7.2 Secondary & Exploratory Endpoints** 1. Participant platelet level results 2. Participant white blood cell count results 3. Participant reticulocytes/normoblast count results 4. Patient-reported impact of luspatercept on Quality of Life (QoL) throughout the treatment period.

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Standard collection via routine Healthcare Professional (HCP) visits and medical record extraction. **Serious Adverse Events (SAEs):** Managed and reported in accordance with local post-marketing pharmacovigilance regulations. **Data Monitoring Committee (DMC):** Internal sponsor review of real-world safety data collected from participating Taiwanese sites.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Real-World Evidence (RWE) cohort (all enrolled, eligible patients receiving  $\geq 1$  dose of luspatercept). **Sample Size Justification:** Target enrollment of  $N=61$  based on the estimated eligible adult patient population with transfusion-dependent  $\beta$ -thalassemia major in the participating Taiwanese centers. **Handling of Missing Data:** Standard observational study handling and imputation methods suitable for incomplete real-world medical records.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study is conducted in accordance with Good Pharmacoepidemiology Practices (GPP), local Taiwanese healthcare regulations, and the Declaration of Helsinki. **Monitoring Plan:** Real-world data abstraction and source data verification will be performed as defined by the observational study protocol. **Audit Trail:** Data entry via a designated Electronic Data Capture (EDC) system maintaining a full audit trail for prospective and retrospective patient data.

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## 11. Study Identifiers & Governance

**Primary ID:** CA056-1132 **Registry Number:** NCT06596642 **Posting ID:** 734905597602233248 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Real-World Study of  $\beta$ -Thalassemia Major Treatment With Luspatercept in Taiwan **Official Regulatory Title:** Taiwan  $\beta$ -Thalassemia Major Real-World Study for Luspatercept

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## 12. Clinical Framework (Thalassemia Specific)

**Primary Condition:** Transfusion-dependent  $\beta$ -Thalassemia Major (Preferred Term: beta-Thalassemia Major) **Disease / Syndrome Definition:** An inherited blood disorder characterized by a decreased synthesis of one of the polypeptide chains that form hemoglobin. Anemia results from this abnormal hemoglobin formation. **Ontology Codes:** MeSH: D013789 | SNOMED ID: 40108008 **Diagnosis Threshold:** Transfusion burden  $\geq 24$  RBC units in 24 weeks without  $\geq 35$ -day transfusion-free period **Medical Methodology:** Routine clinical practice / Luspatercept therapy **Optimization Target:** Reduction of RBC transfusion burden and improvement in QoL

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## 13. Study Procedures & Methodology

**Management Pathway:** Standard real-world clinical pathway for  $\beta$ -thalassemia major management **Education Protocol (HCP):** Local approved label and prescribing guidelines for luspatercept **Education Protocol (Patient):** Standard counseling on potential adverse effects, transfusion monitoring, and QoL survey completion **Quality Audit Mechanism:** Real-world data verification, patient registry cross-checks, and source record reviews

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## 14. Observation & Follow-Up Schedule

**Total Duration:** Observation spanning from baseline up to the end of the study in October 2026 **Onsite Visit Cadence:** Dictated by routine clinical practice and patient needs **Remote Monitoring Frequency:** Data abstraction at predefined intervals throughout the study **Checklist Review Interval:** Routine evaluation of transfusion records and QoL questionnaires

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## 15. Sign-off & Approval

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|--------------------------|--------|-----------|---------------|-----|------|-----|-----|
| Role                     | Name   | Signature | Date          | :-- | :--  | :-- | :-- |
| Principal Investigator   | [Name] | _____     | [DD-MMM-YYYY] |     |      |     |     |
| Clinical Operations Lead | [Name] | _____     | [DD-MMM-YYYY] |     | Lead |     |     |
| Statistician             | [Name] | _____     | [DD-MMM-YYYY] |     |      |     |     |